BRIDGE NAME: _____ BRIDGE NUMBER: _____

JOB NUMBERS: _____

IDENTIFICATION DETAILS: _____

_____

_____

---

DESCRIPTION OF NON-CONFORMITY:

_____

_____

_____

_____

_____ 20____ _____
Date Signature

---

CORRECTIVE ACTION:

_____

_____

_____

_____

_____

ACTION AGREED UPON: _____ _____
Quality Control Supervisor Quality Assurance Inspector

---

DATE COMPLETED: _____ 20 ___ _____
Signature

HOLD REMOVED BY: _____ _____ 20___
Date

QUALITY CONTROL SUPERVISOR: _____ _____ 20___
Date

QUALITY ASSURANCE INSPECTOR: _____ _____ 20___
Date

## SAMPLE FORM #3

### Steel Fabrication Compliance Check List

BRIDGE NAME: _____ BRIDGE NO. _____

MARK NUMBER/S: _____

| | | Q.C. | Q.A. |
|---|---|---|---|
| Material: | Mill Certificates | ____ | _____ |
| | (Physicals, Chemicals, Impact Category) | ____ | _____ |
| | Identification on materials check | ____ | _____ |
| | Heat number transferred and recorded | ____ | _____ |
| Welding: | Procedures approved for each process | ____ | _____ |
| | Consumables checked | ____ | _____ |
| | Date Completed | ____ | _____ |

## SAMPLE FORM #4

### Steel Fabrication Compliance Check List

BRIDGE NAME: _____ BRIDGE NO. _____

MARK NUMBER/S: _____

| | | Q.C. | Q.A. |
|---|---|---|---|
| Welding: | Weld preparation and fit up | _____ | _____ |
| | Welding within parameters | _____ | _____ |
| | Identification on materials check | _____ | _____ |
| | Shear studs | _____ | _____ |
| | Visual Inspection | _____ | _____ |
| | NDT outside agency results | _____ | _____ |

| | | Q.C. | Actual | Q.A. Verified |
|---|---|---|---|---|
| | | Req'd | | |
| Dimensions: | Flanges Top Width 1 | _____ | _____ | _____ |
| | Width 2 | _____ | _____ | _____ |
| | Flatness | _____ | _____ | _____ |
| | Tilt | _____ | _____ | _____ |
| | Bottom Width 1 | _____ | _____ | _____ |
| | Width 2 | _____ | _____ | _____ |
| | Flatness | _____ | _____ | _____ |
| | Tilt | _____ | _____ | _____ |
| | Web Depth | _____ | _____ | _____ |
| | Section Length | _____ | _____ | _____ |
| | Bearing Ctr. Length | _____ | _____ | _____ |
| | Sole Plate Contract | _____ | _____ | _____ |
| | Camber | _____ | _____ | _____ |
| | Sweep Tolerance (1/1000) | _____ | _____ | _____ |
| | Splicing Camber | _____ | _____ | _____ |
| | Alignment | _____ | _____ | _____ |
| | Holes | _____ | _____ | _____ |
| | Date Completed | _____ | _____ | _____ |

## SAMPLE FORM #5

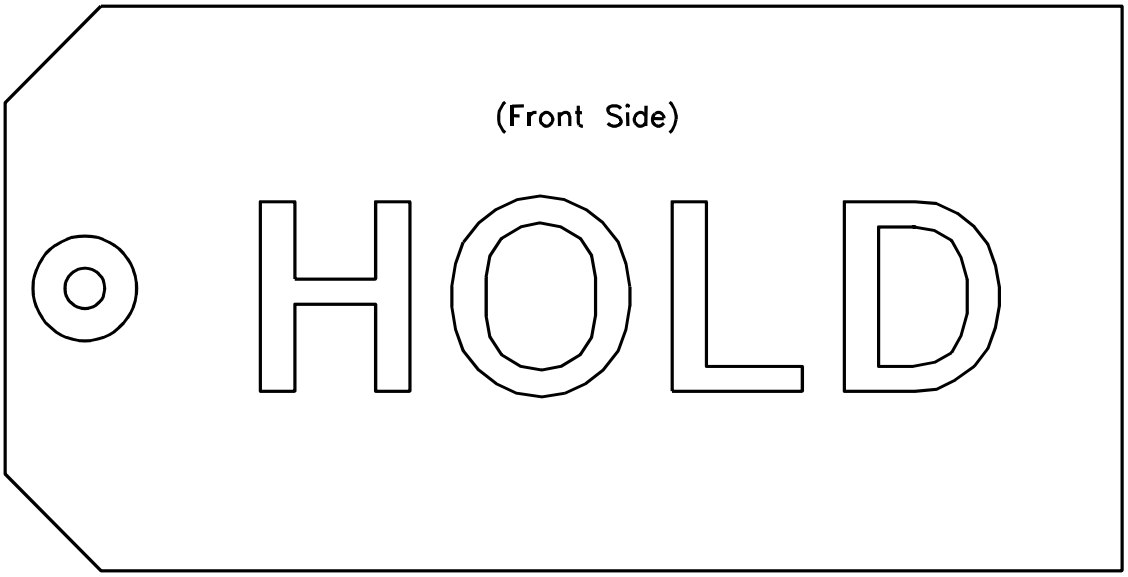
## Coating Compliance Check List

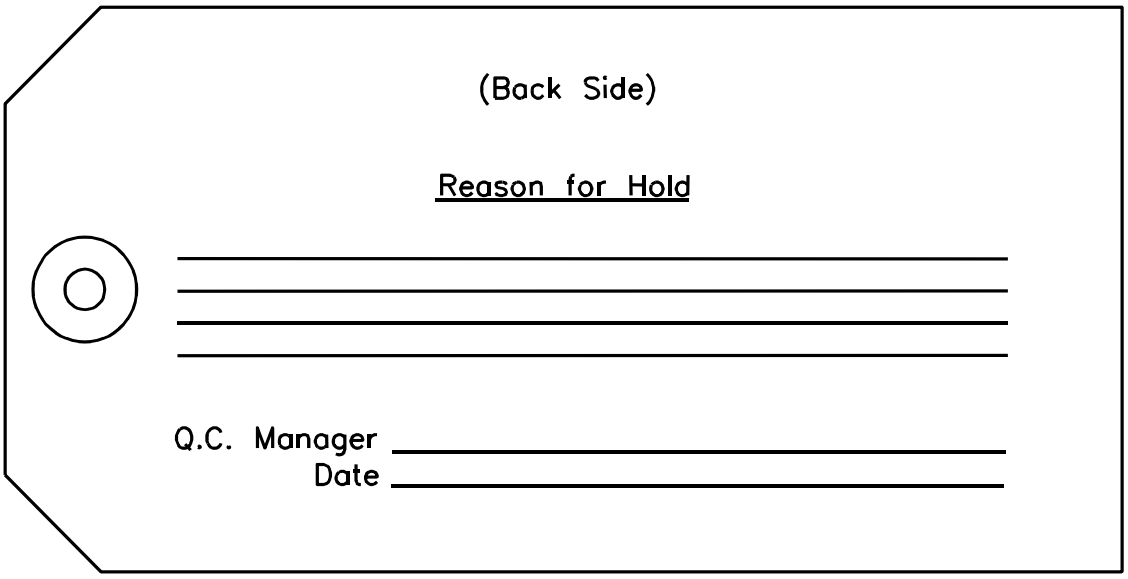
BRIDGE NAME: _____ BRIDGE NO. _____

MARK NUMBER/S: _____

| | | Q.C. | Q.A. |
|---|---|---|---|
| Coating: | WHMIS Safety Data Sheets | _____ | _____ |
| | Technical Spec. Sheets | _____ | _____ |
| | Surface Preparation | _____ | _____ |
| | Ambient Conditions - Req'd | _____ | _____ |
| | - Actual | _____ | _____ |
| | Application | _____ | _____ |
| | Test Results | _____ | _____ |
| | Date Completed | _____ | _____ |

## SAMPLE TAG #6



# SECTION 422

# MISCELLANEOUS STEELWORK

## DESCRIPTION

**422.01 Scope -** This Section covers steelwork apart from major structural steel elements (see DBSS 421).

**422.02 Adherence to Drawings -** Steelwork items shall be fabricated in strict accordance with the Design and the Contract.

**422.03 Working Drawings -** The working drawings shall show all information necessary for the fabrication of the steelwork.

Working drawings shall be in the same system of units as the Design.

Working drawings shall be on D size sheets approximately 560 mm (22") by 865 mm (34"), (one drawing per sheet). Lettering for notes and dimensions shall be at least 2.5 mm and 4 mm for headings. Drawings shall be legible when half-sized or microfilmed.

In the case of simple items, where the Design is complete enough to be used for fabrication, the Contractor may, with approval of the Quality Manager and Ministry Representative, dispense with working drawings. The Contractor shall notify the Quality Manager and Ministry Representative of such intention in writing.

In addition to the drawings submitted to the Quality Manager, the Contractor shall submit to the Ministry Representative four sets of all working drawings, for the Ministry's information only, two weeks prior to the fabrication of the steelwork.

**422.04 Inspection -** The Contractor shall notify the Quality Manager and Ministry Representative at least 14 days before fabrication. The Contractor shall allow the Quality Manager and Ministry's Representative's access to all parts of the work, and shall supply such information and assistance as is required. When the Ministry requests, the Contractor shall provide samples of any materials. Review by the Ministry shall not relieve the Contractor from obligation to perform the work in accordance with the Contract.

Any welding work found to be unacceptable shall be corrected in accordance with CSA W59-M, Section 5.10.

**422.05 Rejections -** The Quality Manager or Ministry Representative may reject any items which do not comply with the requirements of the Contract.

## MATERIALS

**422.11 Materials -** Structural steel, except where shown otherwise on the drawings, shall conform to CAN/CSA-G.40.21-M, Grade 300W or better.

Chemical composition of steel to be galvanized shall be in accordance with DBSS 422.36 – Galvanizing.

The quality and care of electrodes shall conform to the requirements of the latest CSA W59-M.

**422.12 Storage of Materials -** Structural material shall be stored above ground. It shall be kept free from dirt and other foreign matter, and shall be protected as far as practicable from corrosion. Long members shall be supported on skids placed near enough together to prevent injury from deflection.

## CONSTRUCTION

**422.31 Welding -** All welding shall be done in accordance with procedures approved by the Canadian Welding Bureau (CWB) and reviewed by the Quality Manager. All welded fabrication shall be done to the requirements of the latest CSA W59-M.

Welding shall be undertaken by a company approved by the CWB to the requirements of CSA W47.1, Division 3 or better.

**422.32 Welders and Welding Operators -** The Contractor shall produce evidence that all welders and welding operators to be employed on the work are currently qualified by CWB in the processes in which they are to be employed.

**422.33 Flame Cutting -** Steel may be flame cut, using a mechanical guide, provided a smooth surface is secured. Flame cutting by hand shall be done only where approved by the Designer and Quality Manager, and the surface shall be made smooth by planing, chipping or grinding. The cutting flame shall be so adjusted and manipulated as to avoid cutting beyond the prescribed lines. Re-entrant cuts shall be filleted to a radius of not less than 10 mm. Edges of tension members shall be ground to a radius of 2 mm.

All exposed corners of members, which are to be painted, shall be ground to a radius of 2 mm.

**422.34 Bent Plates -** Plates to be bent shall be taken from the stock plates so that the bend line will be at right angles to the direction of rolling. The inside radius of the bend shall not be less than the thickness of the plates. Before bending, the corners of the plate shall be ground to a radius of 2 mm throughout the part to be bent.

**422.35 Shop Painting -** This subsection applies to those parts of the steelwork which are to be painted.

Coating work shall be in accordance with DBSS 216 unless noted otherwise in DBSS 422.

All steelwork, which is to be painted, shall be given three shop coats of paint - primer, stripe coat and midcoat. Paint shall be chosen from DBSS 308 System SS1. Paint shall be supplied by the Contractor.

The topcoat may be applied in the shop or in the field. The

topcoat coat shall be chosen from either DBSS 308 System SS1 or SF2. If the topcoat is to be applied in the shop, then the Quality Management Program shall specifically address the integrity of the topcoat through to project completion.

Faying surfaces shall receive the prime coat only, as per DBSS 421.31.06. Faying surfaces shall not be coated with the stripe coat, midcoat or topcoat paint.

Paint shall be applied in a covered area in accordance with the manufacturer's specifications and DBSS 216.

All edges, corners, crevices, bolts, nuts, protrusions and welds (unless ground flush) shall be stripe painted by brush for a width of 50 mm with the midcoat paint before the midcoat coat is applied. Stripe coating paint shall be applied as per SSPC-PA1 and may be applied by spray, but shall be brushed in. Stripe painting shall be allowed to cure before the midcoat coat is applied.

Surfaces shall be cleaned to "near-white" per SSPC-SP10 / NACE No. 2 and 50 $\mu$m to 75 $\mu$m (2 to 3 mils) sharp/angular profile.

The midcoat shall be applied to all surfaces including tops and sides of the top flanges, except faying surfaces. The midcoat shall not be applied until the primer and stripe coat are accepted by the Quality Manager to be sufficiently cured.

Any surfaces inaccessible after erection, except faying surfaces, shall be given, in addition to the three shop coats, one coat of the topcoat paint appropriate for the paint system being used.

If the topcoat is to be applied in the shop, then the topcoat shall be applied after the midcoat is accepted by the Quality Manager to be sufficiently cured.

The colour of the topcoat will be selected by the Designer. The proposed colour shall be subject to the acceptance of the Ministry Representative.

**422.36 Galvanizing (if required) -** All steelwork to be galvanized shall be galvanized after complete fabrication to the requirements of CAN/CSA-G164-M. The galvanizer shall safeguard against embrittlement as required in CAN/CSA-G164-M, Appendix A. Galvanized members shall be subject, at the discretion of the Quality Manager or the Ministry Representative, to the tests for embrittlement outlined in CAN/CSA-G164-M, Section 5.5.

The chemical composition of steel being galvanized shall be as follows:

- Carbon less than 0.25%
- Phosphorus less than 0.05%
- Manganese less than 1.35%
- Silicon less than 0.03% or between 0.15% and 0.25%

For steel not meeting these chemical composition requirements, special galvanizing techniques shall be developed by the galvanizer to ensure that the specified coating thickness and adherence is achieved. A detailed description of the special techniques shall be submitted to the Designer and Ministry Representative for review 2 weeks prior to galvanizing.

**422.37 Metallizing (if required) -** All surfaces to be metallized shall be treated after complete fabrication in accordance with CSA G189. The zinc coating shall not be less than 0.3 mm in thickness.

**422.38 Installation -** Steelwork shall be installed as shown on the Design.

Railings and fences shall be adjusted to produce uniform height and smooth alignment.

Field welding will be permitted only as shown on the Design.

**422.39 Field Painting (if required) -** This section applies to those parts of the steelwork which are to be painted .

Coating work shall be in accordance with DBSS 216 unless noted otherwise in DBSS 422.

All steelwork shall be thoroughly cleaned of all rust, dirt, dust, oil and other foreign materials. Non-visible salts shall meet the SSPC-SP 12 / NACE No. 5 NV-2 levels. The shop coats of paint shall be touched up as necessary.

**422.39.01 Field Touch-up -** Bare, rusty or damaged areas shall be cleaned to SSPC-SP11, Power Tool Cleaning to Bare Metal. Feather edges into the existing coating and build the coating as per DBSS 216. Build the coating using coatings from the specified DBSS 308 System. Coating system shall be from the same manufacturer as the shop coating system on the steel.

**422.39.02 Topcoat -** The topcoat coat shall be chosen from either DBSS 308 System SS1 or SF2.

If the topcoat has not been applied in the shop, then the steelwork shall be given one topcoat coat of paint applied in accordance with the manufacturer's specifications and DBSS 216.

The colour of the finish coat will be selected by the Designer. The proposed colour shall be subject to the acceptance of the Ministry Representative.

**422.40 Touch-Up of Galvanizing and Metallizing -** All field welds and other damage in galvanized and metallized coatings shall be touched up as follows:

If the contract documents call for touch-up by metallizing, the damaged areas shall be locally sandblasted to "near white" per SSPC-SP10. All sandblasted areas shall be metallized in accordance with CSA G189, to provide a zinc coating not less than 0.3 mm in thickness.

Otherwise, the damaged areas shall be thoroughly cleaned, and painted with two coats of Ministry-approved organic zinc-rich paint.

**422.41 Clean-up -** At the completion of the contract, all miscellaneous steelwork shall be cleaned of concrete spatter, mud, oil and other foreign materials.

# SECTION 501

# HOT MIXED ASPHALT PAVEMENT

**Note:** For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

## PART A - GENERAL

**501.01 Preliminary and General -** This Section describes the materials, equipment and quality of work required for the construction of hot mixed, hot laid asphalt pavement.

**501.02 General Description of Work -** Generally, the work consists of:

**a)** Supplying, screening, crushing or any other method of processing, and improving aggregate to produce asphalt mix aggregate.

**b)** Priming or tack coating the prepared surface.

**c)** Heating the asphalt mix aggregate and mixing it with asphalt cement to produce asphalt mix.

**d)** Hauling, placing and finishing the asphalt mix to produce a high quality asphalt pavement.

**501.03 Asphalt Pavement Classifications -** The Class of asphalt pavement required under the contract will be stated in the Special Provisions and will be generally designated as follows:

- Class 1 - Highways for Heavy Traffic
- Class 2 - General Highways

Pavement Classification as Determined by ESAL

Class 1 pavements shall be utilized where the estimated Equivalent Single Axle Load (ESAL) is greater than 1,000,000 based on a twenty year life span of the pavement structure and an average 4% annual traffic growth rate.

Class 2 pavements shall be utilized where the ESALs are less than 1,000,000 based on the same premise.

## PART B - MATERIALS

All types of hot mix material produced under this section shall be paver laid and of high quality and carefully controlled.

Asphalt mix referred to in this text means any and all of the asphalt cement bound mixes listed in this section.

**501.11 Definitions**

**501.11.01 Hot Mix Asphalt Cement -** penetration grade asphalt used in the asphalt pavement.

**501.11.02 Cutback Asphalt -** penetration grade asphalt which has been blended with light petroleum distillates.

**501.11.03 Emulsified Asphalt -** penetration grade asphalt that has been blended with water and emulsifying agents to form aqueous emulsions, including anionic type, cationic type and High Float type.

**501.11.04 Aggregate -** the crushed or screened gravel.

**501.11.05 Asphalt Mix Aggregate -** the processed crushed aggregate just prior to the addition of asphalt cement.

**501.11.06 Asphalt Base Course Mix Aggregate -** the processed crushed 25 mm minus aggregate just prior to the addition of asphalt cement.

**501.11.07 Recycled Asphalt Mix Aggregate -** the processed crushed aggregate for blending with reclaimed asphalt pavement just prior to the addition of asphalt cement and reclaimed asphalt pavement.

**501.11.08 Supplementary Aggregate** - a crushed or uncrushed aggregate of specified gradation, which, when added to aggregate meeting the requirements of DBSS 501.15.01 through DBSS 501.15.03 will produce asphalt mix aggregate required by the asphalt mix design.

**501.11.09 Asphalt Mix (AM)** - hot plant mixture of asphalt cement and dense graded high quality aggregate.

**501.11.10 Asphalt Levelling Course (LC) -** asphalt mix used to improve crossfall, level, and strengthen existing pavements.

**501.11.11 Asphalt Pavement (AP) -** paver laid asphalt mix compacted to uniform density.

**501.11.12 Additives -** solid or liquid materials to enhance the properties of the liquid asphalt cement or mix.

**501.11.13 Reclaimed Asphalt Pavement (RAP) -** asphalt pavement that has been milled with a cold milling machine.

**501.11.14 Asphalt Base Course Mix (ABC)** - a hot plant mix of asphalt cement and asphalt base course aggregate with lower asphalt cement content than conventional mixes.

**501.11.15 Recycled Asphalt Mix (RAM) -** a carefully controlled, hot plant mix of asphalt cement, graded high quality aggregate and reclaimed asphalt pavement.

**501.11.16 Recycled Asphalt Base Course Mix (RABC) -** a hot plant mix of asphalt cement, reclaimed asphalt pavement and asphalt base course aggregate.

**501.11.17 Reject Mix -** mix that is deemed unacceptable for use in the project by the Quality Manager or Ministry Representative.

**501.12 Supply and Handling of Asphalt Materials**

**501.12.01 Asphalt Cement and Liquid Asphalt Cutback and Emulsion -**, the Contractor will supply the asphalt cement and liquid asphalts required for the work.

**501.12.02 Labour and Equipment -** The Contractor shall supply all labour and equipment to remove asphalt cement and liquid asphalts from tank trucks

**501.13 Standard Tests**

**501.13.01 Asphalt Mix Aggregates -** The asphalt mix aggregates will be tested in accordance with the listed tests in Table 501-A.

## TABLE 501-A TESTS FOR ALL ASPHALT MIX AGGREGATES AND SUPPLEMENTARY AGGREGATES

| ASTM TEST DESIGNATION | TITLE OF TEST |
|---|---|
| C - 136 | Sieve Analysis (Dry) |
| C - 117 | Sieve Analysis (Wash) |
| C - 127 | Relative Density & Absorption - Coarse Aggregate |
| C - 128 | Relative Density & Absorption - Fine Aggregate |
| C - 88 | Aggregate Soundness (Magnesium Sulphate) Tests Granular Surfacing & Base Aggregates |
| D – 2216<br>C - 566 | Moisture Content Determination |
| C-142 | Clay Lumps and Friable Particles |
| D - 2419 | Sand Equivalent Test |
| D6928-03 | Micro Deval Abrasion Loss Test |
| D - 421 | Preparation of Samples for Atterberg Limit |
| D - 4318 | Liquid Limit |
| D - 4318 | Plastic Limit |
| D - 5821 | Fracture Count for Coarse Aggregate - by mass - retained on the 4.75 mm sieve |

**501.13.02 Asphalt Mix and Asphalt Pavement -** The asphalt mix and asphalt pavement will be tested in accordance with the listed tests in Table 501-B.

## TABLE 501-B TESTS FOR ASPHALT MIX AND ASPHALT PAVEMENT

| ASTM | TITLE OF TEST |
|---|---|
| D - 2172 | Quantitative Extraction of Asphalt from Bituminous Mixes |
| D - 1188 | Degree of Compaction of Insitu Asphalt Mix Using Cored Samples |
| D - 2950 | Degree of Compaction of Insitu Asphalt Mix by Nuclear Method |

**501.13.03 Granular Shoulder Materials and Constructed Shoulders -** The granular shoulder materials and the constructed shoulders will be tested in accordance with the listed tests in Table 501-C.

## TABLE 501-C TESTS FOR GRANULAR SHOULDER MATERIALS AND CONSTRUCTED SHOULDERS

| ASTM | TITLE OF TEST |
|---|---|
| D - 698 | Moisture Density Relation Determination, Standard Proctor |
| D - 2922<br>D - 3017 | Field Density Test using Nuclear Gauge |

**501.14 Asphalt Mix Aggregate Requirements -** The crushing of all paving aggregate shall conform to the requirements of DBSS 202.09.

Asphalt mix aggregate shall have a maximum Micro Duval abrasion loss factor of 18 when tested in accordance with ASTM D6928-03.

**501.14.01 Coarse Aggregates:**

**i)** shall be all mineral matter retained on the sieve designated for each individual test.

**ii)** shall consist of crushed stone, crushed gravel, or combination thereof, or materials naturally occurring in a fractured condition, or materials naturally occurring of highly angular nature or rough texture.

**iii)** shall be free from coating of clay, silt or other deleterious material, and shall meet the requirements

listed in Table 501-D.

## TABLE 501-D REQUIREMENTS FOR COARSE AGGREGATES

| REQUIREMENTS | PAVEMENT CLASSES | |
|---|---|---|
| | 1 | 2 |
| Maximum water absorption: % by mass | 2 | 2 |
| Maximum % by mass of clay balls and friable particles | 1.0 | 1.5 |
| 2 Fractured Faces: Minimum % by Mass retained on the 4.75 mm sieve | 85 | 70 |

**501.14.02 Fine Aggregate:**

**i)** shall be all mineral matter passing the sieve designated for each individual test.

**ii)** shall be clean, tough, durable, moderately sharp, and free from coatings of clay, silt, or other deleterious material, and shall contain no clay balls or other aggregations of fine material.

**iii)** Not Used

**iv)** shall have a sand equivalent value of not less than 40.

**501.14.03 Mineral Filler and Mineral Dust**

**i)** The Mineral filler shall consist of all mineral matter passing the 0.600 mm sieve and the mineral dust shall consist of all mineral matter passing the 0.075 mm sieve.

**ii)** Mineral filler and mineral dust shall be free from organic matter.

**iii)** Mineral filler shall be non-plastic mineral filler when tested in accordance with ASTM D 4318.

**501.15 Asphalt Mix Design**

**501.15.01 Gradation Limits -** The Contractor shall supply aggregates falling within the grading limits as specified in Table 501-E. .

**501.15.02 Gradation Refinements -** If the gradation of the aggregate supplied by the Contractor does not fall within the gradation limits specified, such refinements will be made to the gradation as are necessary to comply with the said gradation limits at the Contractor's expense.

**501.15.03 Grading Curve -** Once the grading curve is established within the above limits, the maximum permissible variation of the mean of any four consecutive

## TABLE 501-E GRADATION LIMITS FOR AGGREGATES

| SIEVE SIZE (mm) | PERCENTAGE BY MASS PASSING | | | | |
|---|---|---|---|---|---|
| | COARSE ASPHALT MIX | MEDIUM ASPHALT MIX | FINE ASPHALT MIX | ASPHALT BASE COURSE MIX | ASPHALT BOUND OPEN GRADED BASE MIX |
| 37.5 | 100 | --- | --- | --- | --- |
| 25.0 | 80 - 100 | --- | --- | 100 | 100 |
| 19.0 | 60 - 92 | 100 | --- | 80 - 94 | 75 - 100 |
| 12.5 | 50 - 85 | 84 - 95 | 100 | --- | --- |
| 9.50 | 40 - 80 | 73 - 90 | 90 - 100 | 50 - 84 | 30 - 60 |
| 4.75 | 30 - 65 | 50 - 75 | 55 - 80 | 25 - 55 | 5 - 30 |
| 2.36 | 20 - 50 | 35 - 57 | 32 - 64 | 20 - 45 | 0 - 10 |
| 1.18 | 15 - 35 | 26 - 45 | 24 - 51 | 15 - 35 | --- |
| 0.600 | 8 - 30 | 18 - 34 | 17 - 40 | --- | --- |
| 0.300 | 6 - 22 | 10 - 26 | 13 - 29 | 5 - 20 | 0 - 8 |
| 0.150 | 3 - 15 | 6 - 17 | 8 - 18 | --- | --- |
| 0.075 | 1 - 7 | 3 - 7 | 4 - 10 | 0 - 5 | 0 - 4 |

tests from the grading curve shall be within the tolerance specified in Table 501-F.

## TABLE 501-F MAXIMUM PERMISSIBLE TOLERANCE OF AGGREGATE FROM THE GRADING CURVE

| SIEVE SIZE(mm) | MAXIMUM PERMISSIBLE TOLERANCE % PASSING BY MASS (TEST BCH I - 1 AND 2) |
|---|---|
| 25.0 and larger | ± 5.0 |
| 4.75 to 19.0 | ± 4.5 |
| 1.18 and 2.36 | ± 4.0 |
| 0.600 | ± 3.5 |
| 0.300 | ± 2.5 |
| 0.150 | ± 1.5 |
| 0.075 | ± 1.0 |

These tolerances do not waive the requirement that the running average of four consecutive tests must be maintained at all times inside the limits specified in DBSS 501.15.01. Should the Contractor be unable to maintain the grading curve within the limits specified in DBSS 501.15.01, the Contractor shall, at the Contractor's cost, carry out refinements as required to shift the grading curve away from the limits of DBSS 501.15.01 so as to ensure that a variation cushion, as specified by the maximum permissible tolerances shown in Table 501-F, is effected within the limits of DBSS 501.15.01.

**501.15.04 Asphalt Mix Design -** The Contractor will carry out the asphalt mix design with the aggregate supplied by the Contractor.

The asphalt mix design will establish the design grading curve which will define the proportional amounts of coarse, fine and supplementary aggregates required in the mix.

**501.15.05 Not Used**

**501.15.06 Requirements -** All ingredients supplied by the Contractor for incorporation into the asphalt pavement shall meet all the applicable requirements of DBSS 501 and DB SS 951.

The asphalt mix design, meeting the appropriate Marshall Design requirements of Appendix A, will be performed by the Contractor.

The Contractor shall provide the Ministry Representative with the following at least seven days prior to commencement of the paving:

**i)** , Type of aggregate, source and stockpile location.

**ii)** Asphalt cement source showing type and grading classification information supplied by the refinery. **iii)** A stockpile gradation showing daily test results and stockpile average

**501.16 Asphalt Mix Requirements**

**501.16.01 Test - Asphalt Mix Aggregate -** During all phases of the construction of asphalt pavement, the gradation of the asphalt mix aggregate will be determined.

**i)** Samples for this test can be obtained at the asphalt plant prior to addition of the asphalt cement, or after extraction of the asphalt cement from the asphalt mix, or obtained from samples taken at the roadway before or after compaction.

**ii)** Not Used

**501.16.02 Test - Asphalt Mix -** The asphalt mix will be tested and the measured asphalt cement content shall not vary from the design asphalt content by more than 0.3 kg/100 kg of aggregate.

**501.16.03 Mixing Temperature -** The mixing temperature of the asphalt mix will be selected by the Contractor. . (It will normally be within the temperature range which corresponds to a kinematic viscosity of 150 to 300 $mm^2/s$ of the asphalt cement.)

**501.16.04 Temperature Behind Screed -** The asphalt mix temperature immediately behind the screed shall not be more than 15°C lower than the mixing temperature.

**501.16.05 Residual Moisture Content -** The residual moisture content of the asphalt mix taken in the loaded truck, at the point of loading, shall not exceed 1% by mass. Should foaming or bubbling persist, even below the allowed percentage of moisture, the Contractor shall modify the operations accordingly.

**501.17 Asphalt Pavement**

**501.17.01 In-place Density -** The in-place density of the asphalt pavement will be obtained at random locations and the Degree of Compaction will be computed from the formula:

$$\text{Degree of Compaction} = \frac{\text{In-Place Density of Sample}}{\text{Plant Briquette Density}} \times 100$$

The Degree of Compaction of the mean of any five consecutive tests shall be greater than 97%.

**501.17.02 Smoothness Rating -** The smoothness rating of the finished asphalt pavement will be determined by the Ministry by using a Class 1 profiler and the accurate determination of the International Roughness Rating (IRI) will be measured. A lot is 1 lane km of new pavement (See DBSS 502). The Contractor shall achieve a smoothness of an average minimum value of 1.2 IRI for each lot. The Contractor will have an opportunity to remediate the pavement and the remediation methods shall be approved by the Ministry Representative.

**501.17.03 Finished Pavement -** All finished pavement shall be free from depressions, bumps, or other irregularities exceeding 5 mm over a 3 m straight edge, placed in any direction.

**501.17.04 Requirements -** The pavement shall be homogeneous, free from segregation and shall be uniform with respect to surface texture**501.17.05 Remedial Work -** Areas failing to meet the requirements of Subsections DBSS 501.17.01 through 501.17.04 shall be corrected by the Contractor.. Remedial procedures may be selected by the Contractor subject to approval by the Quality Manager and the Ministry Representative.

The remedial work shall meet all specification requirements and criteria intended for the original work.

## PART C - EQUIPMENT AND PLANT

All equipment, plant and instrumentation shall be capable of performing the work in accordance with this section.

The provisions in DBSS 501.21 to DBSS 501.28 are recommended guidelines only, and as such are not mandatory.

**501.21 Requirements for the Pressure Distributor and Liquid Asphalt Storage**

**501.21.01 Pressure Distributor -** The pressure distributor shall be provided with at least the following equipment:

**i)** A quick opening gate in the dome.

**ii)** An external, readily visible depth gauge, indicating the quantity of asphalt in the tank to the nearest 50 ℓ and:

- a dipstick calibrated in 20 mm increments and a calibration chart showing the volume for each 20 mm of depth, or;
- the manufacturer's calibrated dipstick and chart.

**iii)** A means of heating the asphalt material uniformly to any temperature up to 175°C and maintaining the contents constantly at any selected temperature, without any local overheating.

**iv)** A heavy duty, industrial type thermometer or remote reading dial thermometer.

**v)** Rear mounted spray bars and nozzles set parallel to the surface to be sprayed and capable of vertical and lateral adjustment.

The spray bars to be capable of adjustment to provide a uniformly sprayed surface from a minimum of 0.6 m to a maximum of 6 m.

The spray bars to be capable of producing double coverage at one pass.

The feed to the spray bars to be so arranged that there is a constant circulation of hot asphalt material to the bars to ensure a constant pressure of the liquid asphalt at each nozzle during spraying operations.

The feed manifolds to the spray bars to be provided with adequate filtration to prevent clogging of bars and nozzles.

Spray bar nozzles to be of the same type, size and manufacture and set to produce uniformly fan-shaped sprays without atomization.

Spray nozzles to be provided with valves capable of instant full opening and positive cut-off.

**vi)** A connection to a circulating pressure manifold to which a hose may be attached for a single nozzle hand operated outlet.

**vii)** A pressure pump capable of applying the asphalt material at a minimum rate of 180 ℓ per minute per metre of spray bar.

**viii)** A pressure gauge accurate to within 15 kPa and a meter calibrated in litres per minute, shall be readily visible.

**ix)** A fifth wheel tachometer, calibrated in metres per minute and readily visible.

**501.21.02 Liquid Asphalt Storage -** The Contractor may provide a minimum 40 000 ℓ capacity of on site storage for primer materials.

**501.22 Requirements for all Mixing Plants**

**501.22.01 Uniformity -** Plants to be capable of producing a uniform and homogeneous asphalt mix, as specified.

**501.22.02 Equipment - Preparation of Asphalt Cement**

**i)** The Contractor to supply a minimum of two insulated

tanks for the storage of asphalt cement. The tanks to be equipped with means for heating the material under effective control at all times to the temperatures required, and to have a total minimum storage capacity equivalent to 10 hours of plant actual maximum production capacity.

The tanks o be connected to the circulating system such that each tank can be separately isolated from the system, without interfering with the circulation of asphalt cement to the plant.

Each tank to be provided with its own calibrated dipstick and chart.

**ii)** Asphalt cement to be supplied to the mixing plant by a force feed pump placed at the outlet manifold from the storage tank.

**iii)** The asphalt circulating system to conform to the requirements indicated on SS Drawings SP501-01 through SP501-03.

**iv)** All pipe lines, valve boxes, pumps, meter weigh buckets, other containers or flow lines to be provided with a suitable means of heating or insulation to prevent heat loss and to maintain the asphalt cement at the specified temperature between storage tank and mixer.

**501.22.03 Dust Collectors**

**i)** The Contractor to provide and install such systems of dust collection as may be required to meet all existing regulations.

**ii)** Provisions to be made to re-use the collected material, if required.

**501.22.04 Asphalt Cement Control Unit**

**i)** Satisfactory means to be provided to measure the amount of asphalt cement to be added to the mix with all measuring devices being accurate to ± 2%.

**ii)** A readily accessible totalizer meter to be provided to record the total asphalt cement delivered to the mixer during mixing operations.

**iii)** Mix production from plants equipped with volumetric metering devices to stop whenever the temperature of the asphalt cement passing through the devices fluctuates by more than ± 5°C from the temperature established by the mix Design and for which the metering device has been calibrated. Mixing to only resume when the temperature of the asphalt has been brought within the permissible operating range.

**501.23 Requirements for Batch Plants and Continuous Plants**

**501.23.01 Screens -** Screens to be capable of screening all aggregate to the sizes required for accurate proportioning and o have a designed capacity slightly in excess of the full operating capacity of the mixer.

**501.23.02 Plant Bins**

**i)** The plant to be provided with hot aggregate storage bins of sufficient size to ensure uniform and continuous operation of the plant at its rated capacity. There should be at least three bins. All bins to normally be utilized for gradation control of the aggregate.

A minimum number of two bins may be used if the variation in the crusher gradation analyses conforms to the maximum variation allowances described in DBSS 501.15.03.

The bins to be arranged to ensure separate and adequate storage of the required fractions of aggregate. Each bin to be provided with an overflow chute of such size and at such location as to prevent backing up of material into other bins. Overflow material to be wasted.

**ii)** An adequate additional dry storage bin to be provided for additives when required, and satisfactory provision made for proportioning into the mix.

**iii)** Bins to be provided with electronic type telltales which register on an indicator board plainly visible to the plant operator. When the bins empty to the lower third a red lamp to light up and mixing shall stop immediately.

**iv)** Convenient means to be provided for obtaining truly representative samples from each bin.

**501.23.03 Thermometric Equipment**

**i)** An accurate multipoint continuous recording pyrometer or other approved instrument having a temperature range of 50°C to 200°C subdivided into 2°C graduations to be centrally situated in a protected position where it is readily visible to the plant operator.

**ii)** Premium grade thermocouples or resistance thermometer bulbs to be properly installed and protected at the following reference points:

- In the dryer exhaust manifold between the exhaust fan and the primary dust collector.
- In the dryer discharge chute.
- In the fine aggregate bin, not higher than the bottom quarter point of the bin and as near to the bin discharge point as practicable. The temperature reference point must protrude

sufficiently into the aggregate stream flow and be protected by a small diameter tube to prevent the build up of a coat of fine material.
- In the asphalt circulating main pipe at the nearest elbow or junction to the spray bars or asphalt bucket.
- In each asphalt storage tank, to register the true temperature of the asphalt cement at the level of the tank outlet without sensing temperature effects from the tank heating coils or external conditions.

**501.24 Special Requirements for Batch Plants**

**501.24.01 Plant Scales -** Scales for any weigh box or hopper may be either standard beam or springless dial type accurate to 0.5% of the maximum indicated load.

Scales shall be tested as often as necessary to ensure their accuracy. The Contractor to provide and have at hand a satisfactory number of standard weights for frequent testing of all scales. For each scale, a suitable cradle or platform to be provided for applying the test load so that the load is uniformly distributed. The test weights to be kept clean and stored near the scales.

**501.24.02 Time Lock and Batch Counter -** The mixer to have an accurate time lock to control the operation of a complete mixing cycle by locking the weigh box gate after the charging of the mixer, until the closing of the mixer gate at the completion of the cycle. The mixer gate is to be locked throughout the dry and wet mixing periods.

The dry mixing period is defined as the interval of time between the opening of the weigh box gate and the application of asphalt cement; the wet mixing period is the interval of time between the application of asphalt cement and the opening of the mixer gate.

The control of the timing to be flexible and capable of being set at intervals of not more than two seconds throughout cycles up to three minutes. A mechanical batch counter o be installed as part of the timing device and be designed to register only completely mixed batches.

**501.25 Special Requirements for Continuous Plants**

**501.25.01 Calibration of Asphalt Pumps and Aggregate Feed**

**i)** The plant to provide a means of calibrating gate openings and volumetric pumps by means of mass test samples.

The asphalt mix aggregate fed through individual bin orifices to be capable of being diverted into a suitable test box with separate compartments for each aggregate size.

The test box to be equipped to handle conveniently such test samples varying between 100 kg and 250 kg. The plant to be equipped with an accurate revolution counter mounted on the aggregate feed apron to the mixer.

**ii)** In order to calibrate the volumetric proportioning of asphalt cement, the section of the asphalt flow line between the circulating return valve and the spray bar to be provided with a valve and outlet suitable for bypassing the flow from the volumetric proportioning device into containers capable of holding not less than 2500 $\ell$. The sampling line o be fitted with nozzles equal in type and quantity to the pugmill spray bar. The operating pressure, as registered by the gauge installed upstream of the asphalt meter, for a selected pump setting is not to vary during mixing.

**iii)** A platform scale to be provided by the Contractor for determining the mass of test samples of aggregate and asphalt cement.

**501.25.02 Aggregate - Asphalt Cement Interlock -** Synchronization of aggregate and asphalt cement feeds to be provided to allow interlocking control between the flow of aggregate from the bins and the flow of asphalt cement from the meter or other proportioning source.

**501.25.03 Asphalt Mix Aggregate Sampling -** An efficient and satisfactory method to be provided for obtaining truly representative samples of the total aggregate being fed into the pugmill.

**501.26 Special Requirements for Drum Mixers**

**501.26.01 Plant Generally**

**i)** The drum mixer to be a parallel flow or counter flow type and of approved manufacture.

**ii)** The drum mixer to be used in conjunction with a hot storage silo meeting all requirements of this Section.

**501.26.02 Cold Aggregate Feeders and Bins**

**i)** The Contractor to provide sufficient cold feed bins or feeders to allow for the accurate blending of all materials, including supplemental aggregate(s), required to meet the Marshall Mix Design

**ii)** Each bin to be equipped with a variable speed feeder, adjustable gate and other devices ensuring uniform flow of material.

**iii)** All feeders to be quick adjusting and interlockable and maintain a constant and uniform flow throughout their range of calibration.

**iv)** The total flow of asphalt mix aggregate to be metered by an electronic belt scale interlocked with a variable asphalt feed system, so that the proportions of asphalt cement and aggregate entering the drum remain constant.

**v)** The main feed belt to have a vibrating scalping screen, with openings as required to prevent the feeding of oversize or deleterious material.

**vi)** A readily accessible device shall be provided for obtaining samples from the main feed belt.

**vii)** Feed bins to be equipped with bin level tell-tale lights which are visible to the plant operator. When the bins empty to the lower third, a red lamp to light up and mixing shall stop immediately.

**viii)** The Contractor to provide and have at hand a satisfactory number of standard weights for frequent scale testing.

**501.26.03 Calibration -** The asphalt cement and asphalt mix aggregate feed systems to have provisions to enable easy calibration.

**501.26.04 Monitors, Recorders and Controls -** The drum mixer control console to have at least the following monitors, recorders and controls:

- Individual variable quantity feed belt controls which govern the output from each feed bin.
- Master variable quantity feed control which governs the combined output from the bin feeders in their proper predetermined ratios.
- Belt scale totalizer showing accumulated mass of wet aggregate delivered to the dryer drum mixer.
- Moisture content input control for changing internally the system's wet mass to dry mass.
- Indicator showing the computed total mass per hour of dry aggregate delivered to the drum mixer.
- Control for setting the required percentage of asphalt cement based on the mass of dry asphalt mix aggregate.
- Totalizer for accumulated litres of asphalt cement delivered to the drum.
- Master switch to start and stop the asphalt cement and asphalt mix aggregate feeds.

**501.26.05 Thermometric Equipment**

**i)** An accurate multipoint continuous recording pyrometer or other approved instrument having a temperature range of 50°C to 200°C, subdivided into 2°C graduations, to be centrally situated in a protected position where it is readily visible to the plant operator.

**ii)** Premium grade thermocouples or resistance bulb thermometers to be installed:

- in the asphalt cement circulating main at the nearest elbow or junction to the supply manifold
- in each asphalt cement storage tank to measure the true temperature of the stored asphalt at the level of the tank outlet without sensing temperature effects from the tank heating coils or external conditions
- in the drum discharge chute
- in the hot storage silo, not higher than the bottom quarter point.

**501.27 Requirements for Hot Storage Silos -** Hot storage silos to be enclosed and be of a design that will not cause segregation in the asphalt mix. The conveyor to discharge into an automatically controlled batching device having a minimum capacity of 1.8 t, mounted on top of the hot storage silo. However, should segregation be detected, the Contractor is to modify the operation immediately.

The silo to be equipped with a remote indicator light, visible to the plant operator, indicating when silo is less than one quarter full.

The silo to be set up so as to allow enough room for the haul trucks to pass below the discharge hopper thus allowing direct discharge into the trucks. Loading the haul trucks with the use of a conveyor system will not be allowed.

**501.28 Requirements for Mixing Plants Producing Sulphur Extended Asphalt Mix**

**501.28.01 General -** Equipment normally in contact with sulphur to be thoroughly heat jacketed. All those parts, which may be invaded by sulphur, should also be jacketed.

**501.28.02 Pipe Lines**

**i)** Sulphur lines smaller than 25 mm in diameter are not permitted.

**ii)** Pipe systems should be kept full of liquid sulphur all the time to prevent severe corrosion.

**501.28.03 Valves -** Steam jacketed valves, lubricated or non-lubricated plug valves are accepted. Steam jacketed gate, globe, check and relief valves are also accepted.

A sampling valve to be installed in the emulsion feed line immediately after the blender.

**501.28.04 Pumps -** Sulphur to be supplied to the mixing plant by a positive displacement pump, placed at the outlet manifold from the storage tank.

**501.28.05 Storage Tanks**

**i)** Storage tanks to be insulated. Coils to be able to maintain the sulphur at a temperature of 127°C to 137°C. Temperature indicators and controllers to be installed.

**ii)** Sulphur storage tank to have a minimum storage capacity equivalent to 20 hours of maximum plant capacity.

**iii)** Storage tanks to be grounded.

## PART D - CONSTRUCTION

**501.31 Spray Prime Coat and Tack Coat**

**501.31.01 Prepared Granular Bases and Old Pavements -** All prepared granular bases or old pavements which are to be paved shall be prime coated or tack coated at specified rates.

**501.31.02 Liquid Asphalt - Prime Coat or Tack Coat -** The liquid asphalt for prime coat or tack coat shall be applied when surface and weather conditions are favourable and when the air temperature is above 10°C and rising.

**501.31.03 Tack Coating - Surface -** The surface to be tack coated shall be cleaned of dirt and other foreign material by brooming or other acceptable methods.

**501.31.04 Liquid Asphalt - Spray Temperature -** The liquid asphalt shall be sprayed at a designated temperature within the temperature range which corresponds to a kinematic viscosity of 100 to 150 $mm^2/s$.

**501.31.05 Vertical Surfaces -** The vertical surfaces of roadway appurtenances shall be hand sprayed with liquid asphalt to the top of the new pavement only. Over spraying shall be cleaned. .

**501.31.06 Spray Overlapping -** The liquid asphalt spray

$$E = \frac{\text{Material that Passed the Splitting Screen}}{\text{Total Aggregate Material Smaller than the Splitting Screen}} \times 100$$

shall not be excessively overlapped.

**501.31.07 Spraying Faults -** Any spraying faults resulting from the Contractor's method of operation shall be corrected.

**501.31.08 Excess Liquid Asphalt -** Any excess liquid asphalt remaining unabsorbed shall be blinded with sand or fine aggregate.

**501.31.09 Traffic -** Traffic shall not be permitted on tack coat until it has cured and on prime coat until the primer has been absorbed into the granular surfacing.

Where it is not possible to keep the traffic off the treated surface, the surface shall be blinded with sand or fine aggregate prior to allowing traffic.

**501.31.10 Prime Coat and Tack Coat - Curing -** Normally prime coats shall be allowed to cure for a minimum of 24 hours prior to the placing of asphalt mix. No asphalt mix shall be placed on fresh tack coat that has not had a minimum of 30 minutes to cure.

**501.31.11 Prime Coat - Maintenance -** The Contractor shall, maintain the prime coat and/or tack coat.

**501.31.12 Tack Coat - Application -** Tack coat shall not be applied on sections of roadway longer than can be covered by one day of asphalt plant production.

**501.32 Stockpiling Aggregates**

**501.32.01 Aggregates -** The Contractor shall supply all aggregates for the work from approved sources. Ministry of Transportation pits may be used provided they are developed and worked in accordance with DBSS 202 Part C and the pit development plan. Deviation from the pit development plan will require the approval from the Ministry Representative. No guarantee is given or implied that aggregate from Ministry pits will meet the Specification requirements or provide requisite quantities.

**501.32.02 Not Used**

**501.32.03 Stockpile Construction -** Stockpiles shall be constructed as specified in DBSS 202.21. For drum mixer or continuous or batch plant using cold feed gradation control the Contractor will be required to split the aggregate into coarse and fine stockpiles.

The method and screen size used to split the aggregate will be as follows:

Generally, the screen size separating the coarse from the fines shall be 6.3 mm.

- Screening efficiency is calculated as follows:
- Allowable operating variation in screening efficiency shall be 5%.
- Screening efficiency shall in no case be less than 85%.
- The percentage split for each stockpile shall be provided to the Ministry Representative by the Contractor by the time that 25% of the paving aggregate is placed in stockpile(s). This information is to be provided in writing From the start of production until 25% of the material is in stockpile the contractor shall have a maximum carry-over on the splitting screen of 15% and at no time shall vary by more than ± 2.5%.

**501.32.04 Crushed Aggregates -** Aggregates crushed during the plant-mixing operation shall be stockpiled in such a manner as to avoid their direct feeding into the mixing plant.

**501.32.05 Stockpiles -** Stockpiles of different types of material shall be located and constructed in such a manner as to prevent intermingling of the types.

**501.32.06 Supplementary Aggregate -** Prior to the commencement of the plant mixing the total estimated quantity of supplementary aggregate shall be stockpiled at the plant site.

**501.33 Mixing Plant Operation**

**501.33.01 Mixers**

**i)** With pugmill mixers the asphalt mix aggregate shall be dry mixed for a minimum period of ten seconds. The wet mixing time of the asphalt mix shall be a minimum period of 30 seconds, although this period will be increased, if required, to ensure that a uniform, evenly coated mixture is produced.

**ii)** With drum mixers, the mix retention period shall be dictated by the time required to ensure complete coating.

**501.33.02 Rejection and Disposal -** All asphalt mix failing to meet the requirements of this section will be rejected and shall be disposed of at the Contractor's expense as approved by the Ministry Representative and shall meet all environmental requirements.

**501.33.03 Residual Coating of Particles -** Asphalt mix aggregate particles shall not be coated with residue from improper fuel combustion.

**501.33.04 Lubrication -** Lubrication of conveyor belts with diesel fuel will not be permitted.

**501.34 Hauling the Asphalt Mix**

**501.34.01 Hauling Vehicles - General -** The hauling vehicles shall have tight smooth metal boxes cleaned of all accumulations of asphalt mix and foreign materials. Prior to loading, the truck box may be lightly lubricated with a heat resistant asphalt releasing emulsion, having a maximum silicone content of 1 part per 1000 parts of diluted emulsion. Lubrication with diesel fuel will not be permitted.

**501.34.02 Insulation -** All vertical sides of truck boxes shall be adequately insulated.

All haul vehicles shall be equipped with and use at all times an insulated tarpaulin of such size so as to completely protect the asphalt mix from the weather.

**501.34.03 Mechanical Condition -** Haul vehicles shall be in sound mechanical condition and free from oil and fuel leaks.

**501.34.04 Not Used**

**501.34.05 Delivery -** Asphalt mix shall be delivered to the paver at a constant rate sufficient to allow continuous placement.

**501.35 Placing the Asphalt Mix**

**501.35.01 -** When in the vicinity of an operating paver, workers and inspectors shall stand in full view of the paver operator, clear of side arms and screed, and never in front of the machine hopper.

**501.35.02 -** Asphalt mix shall only be placed on clean dry surfaces free from all foreign materials, when weather and conditions are suitable.

All lifts of asphalt mix shall only be placed on a dry surface when the ambient air temperature is 5$^{o}$C and rising and in addition, the top lift shall not be placed when the average surface temperature is below 10$^{o}$C. The temperature shall be taken in the centre of the travelled lane being paved.

**501.35.03 Pick-up Machines -** When this type of equipment is utilized, the Contractor will only place the equivalent of two tandem loads of mix on the roadway at any given time.

**501.35.04 Asphalt Levelling Course -** Prior to the placement of an asphalt pavement overlay, sections of the existing pavement may require a levelling course of asphalt mix to correct superelevation and to level the roadway as required.

**501.35.05 Asphalt Levelling Course - Requirements -** Asphalt levelling course shall be the class of asphalt pavement as stated in the Contract and shall meet all the requirements of this Section.

**501.35.06 Paving Machine -** The asphalt mix shall be placed by a self-propelled paving machine equipped with an activated screed and strike-off that are heated. The machine shall be capable of spreading the mix to the thicknesses and widths as indicated in the Contract and the Design. A lift thickness shall be not less than 2.0 times nor more than 3.5 times the maximum designated aggregate size of the mix.

**501.35.07 Paving Machine - Equipment -** The paving machine shall be equipped with an automatic grade control device capable of controlling the spreading of the asphalt mix to the desired longitudinal grade and to the designated transverse slope within 1.5 mm/m.

**501.35.08 Speed**

**i)** The speed of the paver shall be adjusted to achieve a continuous lay-down operation producing a smooth homogeneous mat, which meets the requirements of this section. All efforts shall be made to ensure a smooth continuous paving operation with minimal paver stops.

**ii)** When paver speed is in excess of 300 m per hour, the Contractor shall be required to have two qualified joint rakers raking the joint.

**501.35.09 Processing -** During the processing of asphalt mix windrowed in front of pick-up machines the temperature loss between the haul vehicle and the paver hopper shall not exceed 8$^{o}$C in any part of the windrow.

**501.35.10 Placing - Other Methods -** When it is impractical to spread and finish the asphalt mix by paving machine, it may be placed and spread by other methods consistent with prevalent construction practices.

**501.35.11 Hydraulic Screed Extension -** Hydraulic screed extension duplicating the effectiveness of the main screed may be used for narrow pavement widenings. If it is determined that the screed extension is affecting the ride of the finished pavement, it shall be removed from the paving machine.

**501.35.12 Longitudinal Joints -** Longitudinal joints in the top lift of asphalt pavement will only be permitted where lane dividing lines are to be painted.

Longitudinal joints in a lift shall be offset at least 150 mm from joints in the underlying lift.

Longitudinal joints in underlying lifts shall not be offset more than 150 mm from lane dividing lines.

**501.35.13 Traffic -** Traffic shall be minimized over an unmatched longitudinal.

**501.35.14 Longitudinal Edge -** At places where the unprotected longitudinal edge of previously placed mat has been damaged by traffic or equipment it shall be trimmed to provide a standard abutting face against which new mix can be placed and compacted.

**501.35.15 Contact Edge -** The contact edge of abutting cooled pavements shall be thoroughly painted with a thin uniform coat of emulsified asphalt supplied by the Contractor.

**501.35.16 Paving Adjoining Parallel Mats -** When paving an adjoining parallel mat there shall be at least a 50 mm overlap onto the previously paved surface. This overlap shall be properly "raked" to form a homogeneous bond between the two mats.

**501.35.17 Raking** - Any material that is "raked" from the joint shall be placed in a wheelbarrow and deposited and immediately levelled on the shoulder where it can be covered by shoulder aggregate, for all lifts of paving, unless otherwise directed by the Ministry Representative. At no time shall "raked" material be placed in front of the paver.

**501.35.18 Discontinued Paving -** When paving is discontinued in any lane on a traffic bearing roadway, the asphalt pavement shall be tapered down at a maximum slope of 25:1. From the location of the tar paper letdown there shall be at least 1.0 m of pavement that follows the profile and grade of the pavement prior to the start of the 25:1 taper (see SS Drawing SP501-04).

When paving resumes and the tar paper letdown is lifted it shall be loaded out for disposal as it is removed. Disposal shall be in an environmentally sound manner.

When paving resumes, the temporary taper shall be cut back to provide a straight vertical face at the transverse joint, which shall be heated and painted as specified in DBSS 501.35.15.

**501.35.19 Contact Faces -** Contact faces of curbs, gutters, manholes, catchbasins, headers, etc., shall be painted as specified in DBSS 501.35.15 before placing the asphalt pavement.

**501.35.20 Fixed Structures -** After compaction, the pavement surface adjacent to gutters, manholes, headers or other fixed structures shall be slightly higher than the surface of such structures, but never exceeding 6 mm.

**501.36 Compacting the Asphalt Mix**

**501.36.01 Compaction -** The asphalt mix shall be compacted to a minimum density consistent with the requirements of DBSS 501.17.

The Contractor shall supply compaction equipment of specific types to ensure a uniform degree of compaction and surface finish consistent with the requirements of this section. However, one rubber tired roller having a minimum mass of 15 t and capable of providing one complete coverage equal to the rolling width of the machine in one pass, will be required on every project.

Once the compacting effort required to obtain the specified degree of compaction and surface finish is ascertained, the compacting effort and procedure shall not be changed without the written permission of the Quality Manager.

**501.36.02 Equipment -** Compaction equipment shall be capable of producing the required density and surface finish.

**501.36.03 Wheels -** The wheels of compaction equipment in contact with the asphalt mix shall be clean. In addition, compactors shall be equipped with mats.

All wheels shall be moistened by water or by an approved diluted asphalt release agent to prevent pickup

**501.36.04 Operation -** Compaction equipment shall normally be operated with the drive roll or wheels nearest the paving machine.

**501.36.05 Longitudinal Joints -** Longitudinal joints shall be rolled immediately behind the paving operation, after which rolling shall begin longitudinally at the lower edge of the mat proceeding toward the higher edge of the mat overlapping on successive passes by at least one half of the effective roller width.

**501.36.06 Roller Speed -** The speed of the roller shall not

$$\text{Degree of Compaction} = \frac{\text{In-Place Dry Density}}{\text{Maximum Dry Density}} \times 100$$

exceed 8 km/h.

**501.36.07 Rollers -** Rollers shall not be driven on to, or off, the unprotected longitudinal edge of the unfinished mat, nor will rollers be allowed to park on the mat prior to complete cooling.

**501.36.08 Vibrating Tampers and Hand Rollers -** Along curbs, manholes, headers and similar structures which are not accessible to rollers, thorough compaction shall be obtained by use of vibrating tampers and hand rollers.

**501.36.09 Traffic -** Traffic will not be permitted on the finished pavement until it has cooled to such a temperature as to ensure that no deformation or excess flushing of the surface will occur.

**501.37 Shouldering -** The materials, equipment and work practice required for the construction of granular shoulders shall be as follows:

**a)** The gradation of the crushed aggregate supplied for granular shoulders shall be within the limits set for medium mix under DBSS 501.15.01 and DBSS 501.15.02, or as specified by the Ministry Representative, and the aggregate quality shall meet all the requirements of DBSS 202.

The finished surface of the granular shoulder shall be true to grade and cross section, and be free from ruts, ravel, segregation or other irregularities.

**b)** Shouldering machines shall comply with the following minimum requirements:

- The machine shall travel on the adjacent asphalt pavement on pneumatic tires or rubber tracks.
- The machine may be self-propelled, towed or pushed.
- The speed of the aggregate feeder shall be adjustable and the feeder shall have sufficient power to feed required quantities of aggregate.
- The machine shall have an adjustable strike-off and cut-off plate to regulate the depth, width and cross slope of the deposited shoulder aggregate.
- The machine shall be capable of spreading material from widths of 0.5 m to 2.5 m.
- All granular shoulder materials shall be handled in such a manner that segregation does not occur.

**c)** If necessary for compacting, water shall be applied to the granular shoulders.

The in-place dry density of the granular shoulders will be obtained at random locations, after completion of final rolling.

The Degree of Compaction of each in-place density test will be computed from the formula:
The average Degree of Compaction of the granular shoulder shall be greater than 98%.

**d)** If the adjacent asphalt pavement is damaged during the shouldering operation, the damage shall be repaired.

**e)** For safety reasons, when a roadway with gravel shoulders is used by the travelling public, an intermediate layer of shouldering aggregate shall be placed following final rolling of levelling course or multiple lifts of asphalt pavement thicker than 50 mm, in advance of the main shouldering operation.

## APPENDIX A

### 1) MARSHALL DESIGN CRITERIA

**(See DBSS 501.15.06)**

| PROPERTY OF LABORATORY COMPACTED PAVING MIXTURE | A.M. or R.A.M. PAVEMENT CLASS | | A.B.C. or R.A.B.C. PAVEMENT CLASS | |
|---|---|---|---|---|
| | 1 | 2 | 1 | 2 |
| ESALs | $>10^6$ | $<10^6$ | $>10^6$ | $<10^6$ |
| Number of blows each face of test specimens | 75 | 75 | 75 | 75 |
| Minimum % Voids in mineral aggregate for maximum particle size: 25.0 mm | --- | 13 | 13 | 13 |
| Minimum % Voids in mineral aggregate for maximum size: 19.0 mm | 14 | 14 | --- | --- |
| Minimum % Voids in mineral aggregate for maximum size: 12.5 mm | 14 | 14 | --- | --- |
| Minimum % Voids in mineral aggregate for maximum size: 9.5 mm | 16 | 16 | --- | --- |
| Percentage of air voids in laboratory compacted mixture for: Coarse Mix | --- | 3 to 8 | 4 to 6 | 3 to 8 |
| Percentage of air voids in laboratory compacted mixture for: Medium Mix | 3 to 5 | 3 to 5 | --- | --- |
| Percentage of air voids in laboratory compacted mixture for: Fine Mix | 4 to 6 | 3 to 8 | --- | --- |
| Minimum Marshall Load, N @ $60^oC$ for 80 - 100 | 9000 | 7000 | 8000 | 7000 |
| Minimum Marshall Load, N @ $60^oC$ for 150 - 200 and 200 – 300 | 7000 | 6000 | 7000 | 6000 |
| Flow index, units of 0.25 mm | 8 to 14 | 8 to 16 | 8 to 14 | 8 to 16 |
| Minimum Index of Retained Stability after immersion in water at $60^oC$ for 24 hours | 85% | 75% | --- | --- |

# SECTION 502

# END PRODUCT SPECIFICATION FOR ASPHALT PAVEMENT CONSTRUCTION

## PART A - GENERAL

**502.01 Preliminary and General –** This Section describes the materials, equipment, professional standards and end product requirements for the construction of hot mixed hot laid asphalt pavement.

End Product Specifications contain the acceptance and payment criteria based on the results of specified sampling and testing. Payment for the asphalt pavement product is subject to adjustments upward and downward in accordance to the provisions provided in End Product Specifications (EPS).

**502.02 General Description Of Work -** Generally, the work associated with the construction of asphalt pavement by EPS consists of the following:

- Preparing a Quality Management Plan before commencing the Work and providing at the production site a testing facility to provide the data needed to implement that plan.
- Supplying, screening, crushing, processing and improving aggregate to produce asphalt mix aggregate.
- Supplying and delivering asphalt cement and spray primer meeting the requirements of DBSS 952.
- Preparing mix designs which, once reviewed and accepted by the Quality Manager, become the basis for the accepted Job Mix Formula.
- Heating the asphalt mix aggregate and mixing it with asphalt cement to produce asphalt mix that meets the Job Mix Formula.
- Hauling, placing, compacting and finishing the asphalt mix.

The Contractor shall provide an end product conforming in quality and accuracy of detail to the dimensional and tolerance requirements of the contract. Where no tolerances are specified, the standard of workmanship shall be in accordance with normally accepted good practice and those identified in DBSS 501. Payment is subject to adjustments based on quality acceptance tests performed by the Ministry Representative and calculations performed by the Contractor with respect to application rate.

**502.03 Definitions Relevant To End Product Specification**

**502.03.01 End Product Specification (EPS) -** A specification whereby the Contractor is entirely responsible for quality control of the construction processes, and whereby the Ministry performs the specified quality audit sampling and testing of the end product for the purpose of determining acceptance/rejection and payment adjustment.

**502.03.02 Job Mix Formula -** The Job Mix Formula establishes the aggregate proportions, gradation, and the asphalt content to be used for production of asphalt mix. The Job Mix Formula is based on a reviewed asphalt mix design, in accordance with DBSS 502.05.07, or on an accepted variation in accordance with DBSS 502.05.09. In either case, each Job Mix Formula requires an independent evaluation.

**502.03.03 Asphalt Content -** Asphalt Content means the quantity of asphalt cement in the mix expressed as a percentage by weight of the total dry aggregate in the mix.

**Design Asphalt Content -** The asphalt content upon which the Job Mix Formula is initially established.

**Approved Asphalt Content -** The design asphalt content or subsequent adjustment to it, embodied in a Job Mix Formula or revised Job Mix Formula which has been evaluated in writing by the Ministry Representative.

**Actual Asphalt Content -** Actual asphalt content is the amount of asphalt cement in the mix as determined by the Ministry's Quality Audit Program.

**502.03.04 Smoothness -** Smoothness is a measure of the longitudinal profile of the pavement surface. The unit for measurement is the International Roughness Index (IRI).

**502.03.05 Sample Mean -** The Sample Mean is the arithmetic mean of a set of test results constituting the sample.

**502.03.06 Stratified Random Sample -** A Stratified Random Sample is a set of test measurements taken one each from a number of separate (stratified) areas or Sub-Lots within a Lot in an unbiased way.

**502.03.07 Lot -** A Lot is a portion of the work being considered for acceptance and for the determination of payment. A Lot is defined as follows.

For the application of the contract requirements for:

- Density,
- Asphalt Content,
- Aggregate Gradation, and
- Material Application Rate,

A Lot shall be one day's scheduled production of at least 7 hours plant production where no changes have occurred to criteria such as but not limited to:

- accepted Job Mix Formula,
- the specific lift that is being placed,
- the required material application rate.

A change in any above criteria may require a new Lot designation.

One day's production of less than 7 hours will be dealt with as follows:

- the material will be added to the next Lot that has the same criteria, as described above, except that if a test indicates that this production is subject to a payment adjustment or to rejection, or if no further material will be produced with the same criteria, this production will be designated as a separate Lot,

A Lot shall be no more than two days total production even if the above criteria have not changed or been met.

For application of the contract requirements for:

- segregation, and
- smoothness,

A Lot shall be one kilometre length of top lift pavement for each driving lane.

**502.03.08 Sub-Lots for Density, Asphalt Content, Gradation, and Smoothness -** For the application of the contract requirements for:

- density, asphalt content, and gradation, each Lot shall be divided into three equal Sub-Lots, defined by lineal metres of production.
- smoothness, each Lot shall be divided into 100 metre Sub-Lots.

**502.03.09 Driving Lane -** A driving lane shall mean a single lane in any area of the pavement other than a shoulder or a barrier flare.

**502.03.10 Not Used**

**502.04 Quality Control**

**502.04.01 General -** Quality Management, by the Contractor, is the sum total of activities by the Contractor to ensure that a product meets contract specification requirements. Quality Management includes material handling and construction procedures, calibration and maintenance of equipment, production process control and any sampling, testing and inspection that is done for these purposes.

The Contractor is responsible for all Quality Management under this specification. The Ministry Representative will audit and monitor the Contractor's operations and the implementation of the Contractor's Quality Management Plan.

The Ministry will not take samples for quality control or quality assurance testing and will in no manner assist in any degree or in any aspect of the Contractor's operation in the production of asphalt pavement, beginning from the production of aggregate through the compaction of asphalt mix.

The Ministry Representative may issue a Stop Work Order to the Contractor if the Contractor fails to adhere to the Quality Management Plan. The Ministry Manager will not issue a Resume Work Order until the Contractor has given the Ministry Representative satisfactory assurance that it has in place adequate capacity to fulfil the requirements of the Quality Management Plan.

**502.04.02 Quality Management Plan -** The Contractor shall prepare a detailed written Quality Management Plan that fulfils or exceeds the requirements of Appendix 1 and functions integrally with any other Quality Management provisions of the Contract. The Contractor's Quality Management Plan shall be submitted to the Ministry Representative not less than five working days before the production of paving aggregate is started. The Ministry Representative will review the plan for approval based on whether or not it meets the Quality Management Requirements and Guidelines, and will respond, in writing, within five days. Any subsequent change to the Contractor's Quality Management Plan must also be submitted to the Ministry Representative, for approval, not less than twenty-four hours before such change is implemented.

The Contractor's Quality Management Plan shall include a description and schedule of the intended sampling, testing and reporting. The Quality Management Plan must also include a detailed description of the means by which the Contractor shall use the quality management test results to ensure that the asphalt materials, aggregate, mix production, paving and pavement compaction processes will be controlled to keep the end product within the specified limits. The Quality Management Plan must clearly show the flow of information from the laboratory to the individuals who shall make the actual adjustments to the processes and equipment to affect this control. The plan will show time allowance for each step, the names and positions of all the people involved, and a clear description of the responsibilities of each.

**TABLE 502-A-1 – STANDARD SAMPLING AND TESTING PROCEDURES FOR AGGREGATES**

All aggregate tests shall use the following sieve sizes:
37.5mm, 25mm, 19mm, 16mm, 12.5mm, 9.5mm, 4.75mm, 2.36mm, 1.18mm, 0.600mm, 0.300mm, 0.150mm, and .075mm.

| Item | Procedure | *Test Reference No.* |
|---|---|---|
| 1 | Sieve Analysis of Fine And Coarse Aggregates | ASTM C-136 |
| 2 | Materials Finer Than 0.075 mm (No. 200) Sieve in Mineral Aggregate by Washing | ASTM C-117 |
| 3 | Specific Gravity and Absorption of Coarse Aggregate | ASTM C-127 |
| 4 | Specific Gravity and Absorption of Fine Aggregate | ASTM C-128 |
| 5 | Determining the Percentage of Fractured Particles in Coarse Aggregate | ASTM D-5821 |
| 6 | Total Moisture Content of Aggregate by Drying | ASTM C-566 |
| 7 | Standard Practices for Sampling of Aggregates | ASTM D-75 |

**TABLE 502-A-2 – STANDARD SAMPLING AND TESTING PROCEDURES FOR ASPHALT CEMENT**

| Item | Procedure | *Test Reference No.* |
|---|---|---|
| 1 | Penetration of Bituminous Materials | ASTM D-5 |
| 2 | Viscosity of Asphalt by Vacuum Capillary Viscometer | ASTM D-2171 |

**502.04.03 Quality Management Testing and Inspection -** The Contractor shall provide and maintain equipment and qualified personnel to perform all laboratory testing, field testing and inspection necessary to determine and monitor the characteristics and properties of all the materials produced and incorporated into the work. They shall also monitor the workmanship of the final product in accordance with the Quality Management Plan as most recently submitted.

The Contractor's Quality Management testing and inspection shall be done under the direction of the Quality Manager.

The Contractor shall provide a testing facility(s) that meets the requirements necessary to carry out all the test procedures listed within this Section. The facility(s) must have the equipment specified under the appropriate test designation to perform the tests. The Ministry shall have access at all times to the quality control facility(s).

**502.04.04 Quality Management Records -** The results from quality testing shall be reported on test logs and plotted on charts immediately after each test is completed. The Contractor shall report all test results on Ministry supplied forms available from the Ministry Representative. Reports and charts shall be available for viewing to the Ministry Representative, within 24 hours of the end of each shift.

For the purposes of confirming delivery of asphalt mix to the road and the calculation of material application rates, the Contractor shall consolidate the Road Checker's Summary and the weigh tickets for each load received at the paving operation at the end of each shift. The Road Checker's Summary shall include, but not be limited to, the following information:

**a)** Truck Number

**b)** Weigh Ticket Number and Net Weight of load

**c)** Time and location by station of delivery

**d)** Material Application Rate Dimensions and Calculations. Calculation frequency minimally shall be for every 10 loads.

**e)** Notes pertaining to the paving of any appurtenances (letdowns, intersections, tapers, etc.)

ASTM test procedures are to be followed.

All equipment and the laboratory shall be well maintained and in good working condition. All testing equipment shall be calibrated and evidence of the calibration shall be provided when requested by the Ministry Representative. For the purpose of obtaining accurate and consistent results between the Contractor's Quality Management testing and the Ministry's Quality Audit testing, correlation of the Marshall hammer and ignition ovens to be used on the project shall be performed during the verification of the Job Mix Formula.

**TABLE 502-A-3 – STANDARD SAMPLING AND TESTING PROCEDURES FOR ASPHALT MIX AND PAVEMENT**

| Item | Procedure | *Test Reference* |
|---|---|---|
| 1 | Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures | ASTM D-2726 |
| 2 | Bulk Specific Gravity and Density of Compacted Bituminous Mixtures Using Paraffin Coated Specimens | ASTM D-1188 |
| 3 | Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus | ASTM D-1559 |
| 4 | Effect of Water on Compressive Strength of Compacted Bituminous Mixtures | ASTM D-1075 |
| 5 | Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures | ASTM D-2041 |
| 6 | Percent Air Voids in Compacted Dense and Open Bituminous Paving Mixtures | ASTM D-3203 |
| 7 | Calculating Percent Asphalt Absorption by the Aggregate in an Asphalt Pavement Mixture | ASTM D-4469 |
| 8 | Sampling Compacted Bituminous Mixtures for Laboratory Testing | ASTM D-5361 |
| 9 | Determining Asphalt Content by Ignition Method | ASTM D-6307 |
| 10 | Combining Loose Mix for the purpose of Determining Asphalt Content and Gradation | **APPENDIX 2** |

**TABLE 502-A-4 – TEST SAMPLE SOURCE LOCATIONS**

| **Test** | **Sample Source:** |
|---|---|
| Density | 100mm dia. road cores; 150mm dia. Road cores for Superpave |
| Asphalt Content | Loose Mix Samples |
| Smoothness | **Centre of the Lane** |
| Aggregate Gradation | Loose Mix Samples |
| Segregation | Roadway Pavement |